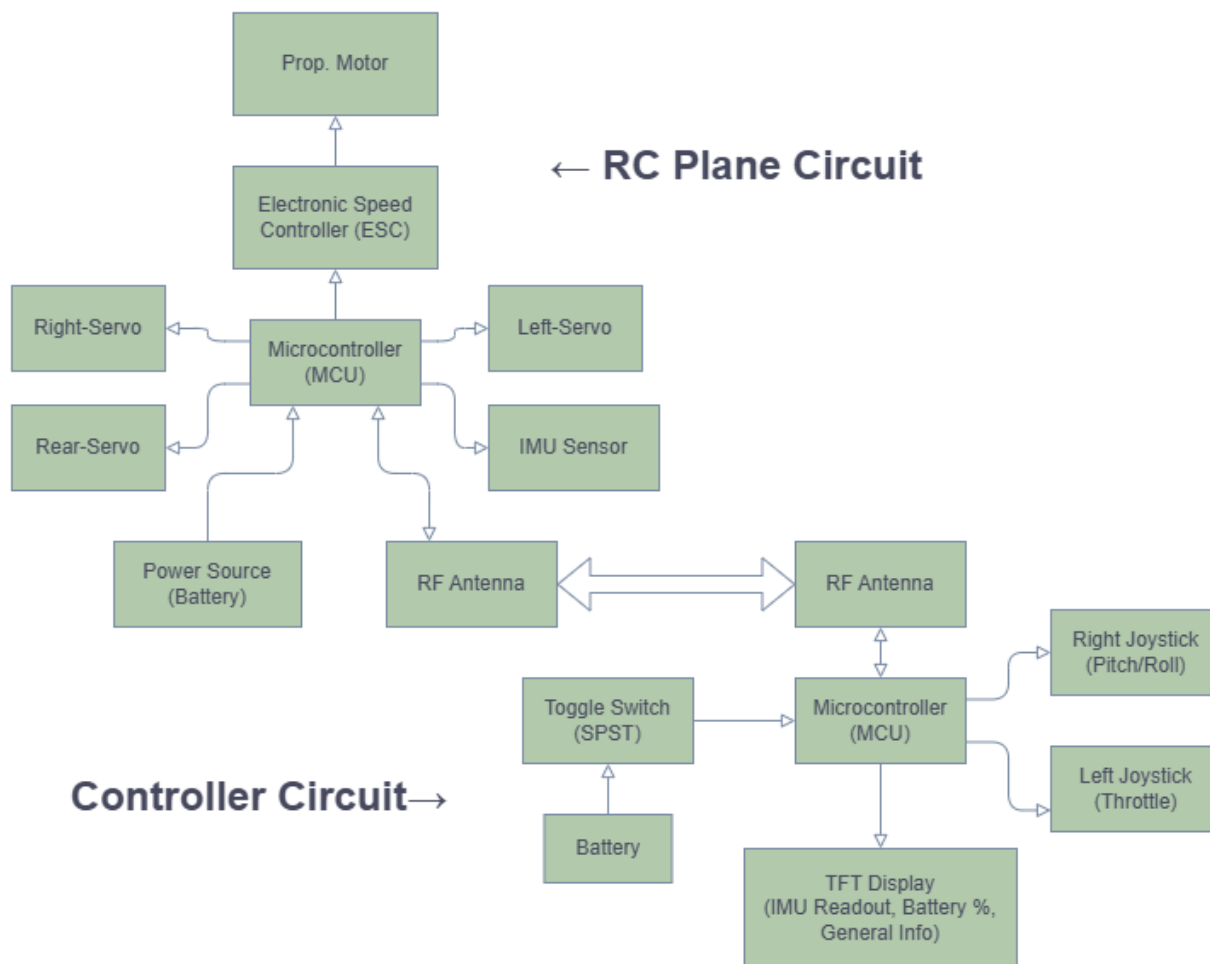


ENGR 2999 Progress Report 1

Project Introduction

For our ENGR2999 Capstone project we will be designing and fabricating a RC airplane that responds to the user interface of a joystick and throttle to control motor power and flap control. Our project block diagram is shown below.



Task Breakdown

Our development process will comprise the following tasks:

1. Component Selection:
 - a. Which microcontroller (MCU) we will be developing around
 - b. Picking our control method between the controller and plane (*RF, IR...*)
 - c. Miscellaneous component selection:
 - i. Servos
 - ii. Control surfaces (*joysticks, buttons...*)
 - iii. Sensors (*gyroscopes, accelerometers, pressure sensors...*)

2. Mechanical Platform Prototyping:
 - a. Material testing and selection
 - b. Modelling and prototype fabrication
 - c. Test fitting and electrical component positioning
 - d. Estimating/Calculating final expected mass, drag and required thrust
 - i. *Allows for propellor motor, electronic speed controller (ESC) and battery selection.*
 - e. Initial circuit testing and prototyping
 - i. Prototyping individual circuit functions
 - ii. Building firmware/program prototype functions
3. Mechatronic Platform Fine Tuning:
 - a. Continuing iteration on the following:
 - i. Mechanical design, weight reduction, and aerodynamic optimization.
 - ii. Compiling program functions into single program and debugging.
 - iii. Preparing full circuit for chassis installation and mechanical integration.
 - b. Verifying component selections and potentially identifying replacements.
 - i. Potential issues such as:
 1. Overweight components
 2. Underpowered actuators
 3. Battery over/under capacity (over capacity is relative to weight)
 4. Antenna troubleshooting and unknowns
 - c. Debugging, debugging, debugging...
4. Final testing:
 - a. Verifying that all systems work
 - b. Reasonably optimized design and code
 - c. Final test flight

Task Schedule

Our team of 2 people will split into two groups with the following flexible division of labour: Brendan Morrison will work on the hardware team. The hardware team will work on:

- Material Selection and Structural Analysis:
 - Evaluate strength-to-weight ratios of different materials.
 - Perform necessary thrust calculations.
- Mechanical design:
 - Develop 2D schematics of interior component packaging and exterior aerodynamic profiles.
 - Utilize Inventor Autodesk software to create a full assembly of airframe.
 - Use inventor model to calculate centre of gravity to ensure flight stability.
- Mechanical assembly:
 - Design custom mounts for motor, servos, and battery tray.
 - Ensure control linkages operate smoothly and efficiently.
- Prototyping:
 - Utilize 3D printing for quick iterations of structural components.
 - Analyze wear and tear after test flights and reinforce components where necessary.

Max Larouche will form the electronics team. The electronics team will work on:

- Selecting electrical components

- Electronics design:
 - High-level schematics for developing early design aspects for the plane and circuit.
 - Low-level schematics for final circuit setup.
- Electronics assembly:
 - Fitting and positioning electrical components in the plane housing.
 - Wiring and soldering the circuit.
- Software development:
 - High-level design for program execution and task management programs.
 - Prototyping individual plane and controller functions.
 - Firmware development and debugging.
- Overall electrical design process:
 - Prototyping individual circuit subsections and program functions.
 - Firmware development and compilation throughout.
 - Circuit design, prototyping, iteration, and final assembly.

